Free pythagorean theorem worksheet worksheet 7

Maths | 10th grade | Mixed Operations

1. If f(x) = 2x - 3 and g(x) = x^2 + 4, find (f o g)(2).

A) 11
B) 5
C) 13
D) 9

2. Solve for x: 3x - 4 = 2x + 5.

A) 9
B) -9
C) 1
D) 0

3. Simplify the expression: 5(2x + 3) - 4x.

A) 10x + 15
B) 6x + 15
C) 10x + 3
D) 6x + 9

4. Evaluate the expression when a = 3 and b = 2: 4a^2 - 2b^3.

A) 36
B) 28
C) 44
D) 46

5. Solve for y: 7y + 8 = 3y - 4.

A) -3
B) 3
C) 1
D) -1

6. What is the solution to 5 - 3(x + 2) = 2?

A) -1
B) 1
C) 0
D) -2

7. Find the value of m if 4m + 6 = 2(m + 8).

A) 5
B) 8
C) 2
D) 0

8. If the sum of three consecutive integers is 18, what are the integers?

A) 5, 6, 7
B) 6, 7, 8

C) 4, 6, 8
D) 4, 5, 6

9. Determine the value of k if the roots of x^2 - 5x + k = 0 are equal.

A) -25
B) 25/4
C) 0
D) 25

10. In a geometric sequence, the third term is 18 and the sixth term is 486. Find the common ratio.

A) 2
B) 3
C) 4
D) 6

Answer Key

1. C
First calculate g(2) = 2^2 + 4 = 8. Then, f(8) = 2(8) - 3 = 13.

2. A
Subtract 2x from both sides: x - 4 = 5. Then, add 4: x = 9.

3. B
Distribute the 5: 10x + 15, then subtract 4x: 6x + 15.

4. B
Calculate: 4(3^2) - 2(2^3) = 4(9) - 2(8) = 36 - 16 = 28.

5. A
Subtract 3y from both sides: 4y + 8 = -4. Then subtract 8: 4y = -12. Divide by 4: y = -3.

6. C
Simplify 5 - 3x - 6 = 2 to -3x - 1 = 2. Add 1: -3x = 3. Divide by -3: x = 0.

7. C
Expand the right side: 4m + 6 = 2m + 16. Subtract 2m: 2m + 6 = 16. Subtract 6: 2m = 10. Divide by 2: m = 5.

8. B
Let the integers be x, x+1, x+2. Their sum: 3x + 3 = 18. Solve: 3x = 15, x = 5. Integers are 5, 6, 7.

9. B
For equal roots, discriminant 9B Ò %ã" Ò F 2 Ò à Here, a=1, b=-5, c=k: (-5)^2 - 4(1)(k) = 0, so 25 - 4k = 0. Therefore, k = 25/4.

10. B
Let the first term be a and the common ratio be r. The sixth term a*r^5 = 486, the third term a*r^2 = 18. Divide the equations: r^3 = 486/18 = 27, thus r = 3.